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RA 10173 and its Challenges to Philippine State Universities and Colleges' Compliance Performance: The Case of Mindanao State University- General Santos City

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ABSTRACT

Data privacy has become a major concern in the Philippines owing to the promulgation of Republic Act No. 10173, otherwise known as the Data Privacy Act of 2012, duly enforced by the National Privacy Commission (NPC). This study seeks to determine the level of compliance of Mindanao State University- General Santos City (MSU-GSC) as well as understand the challenges it encounters in accordance with the Data Privacy Act. Using a qualitative single holistic case study design and pattern matching technique, this paper was able to uncover how and why SUCs can comply with the RA 10173 and can apply such law in their information systems and academic processes. It was found out that MSU-GSC is qualitatively described as partially compliant and it further supports deterrence and legitimacy as its determinants. The moderating effect of compliance with RA 8792 or E-Commerce Act of 2000 also significantly affects its abidance with Data Privacy Act as efforts are continuously directed towards automating university processes and transactions. Moreover, three factors were identified that contributes primarily on the challenges of its compliance namely lack of better understanding, budgetary issues and time constraints. The study contributes to the literature by providing inputs about SUCs' continuing efforts toward compliance with RA 10173. It is suggested NPC to diffuse responsibility to SUCs, so as compliance with data privacy act will weigh down less. Further, it is expected a greater appreciation of the law will fuel enthusiasm to overcome challenges of compliance.

CCS Concepts

Social and professional topics → Computing/Technology policy → Privacy policies

Keywords

Data Privacy; Personal Information; Data Privacy Compliance; e-Governance

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1. INTRODUCTION

1.1 Background of the Study

Data privacy has been defined in many ways by literatures. It has been thought of as a moral or legal right, having control over information about oneself [1] or having control as to how it is acquired and used [2]. In spite simple definitions, its varied nature has earned data privacy its place as one of major concerns in different domains such as information systems, marketing, psychology and management among others [2]. In the Philippine setting, a culture of protecting the data privacy of citizens while keeping a free-flow of information forms part of the government's commitment to serve and defend its people well [3]. Pursuant to the National Privacy Commission's (NPC) Circular Number 16-01, all heads of government branches, bodies or entities including those of state universities and colleges were reminded of the provision of the law that applies to all government agencies that are engaged in the processing of personal data in light of RA 10173 or otherwise known as Data Privacy Act of 2012. The circular affords to assist these government agencies engaged in processing of personal data as the rules stipulated therein can be used to craft detailed policies and procedures along the line of their operating requirements.

As information technology continues to change the way we do transactions on a daily basis, concerns about data protection and usage are put in the limelight [4]. As SUCs continue to invest on data systems to improve quality and increase proficiency in managing data, a clear understanding of and commitment to data privacy and a rational implementation and supervision of strong policies and practices are encouraged to protect student data from misuse [5]. As such, effective use of data is highly recommended. Consequently, a balance between the data's use and protection can be achieved so as to mitigate risks related to its intentional or unintentional manipulation. While the law is strongly enacted, SUCs however are faced with numerous impediments to its compliance owing to the changes it may bring to its operations [6]. In spite these challenges, SUCs continuously pursue abidance along the line of its implementation to meet legal responsibilities and to respect confidentiality of personally identifiable information.

To contribute to the understanding of this phenomenon, this case study research was conducted following a single holistic approach [7], taking into account the context within which abidance with the law is implemented by Mindanao State University- General Santos City (MSU-GSC). Moreover, this paper aims to uncover efforts

and challenges made by MSU-GSC with respect to RA 10173 thereby assessing its level of compliance.

1.2 Overview of Related Literature

1.2.1 Data Privacy in Education Sector

The fast-paced technological adoption of systems that cater to store and analyze academic outcomes at a granular level has become an opportunity as well as threat to the education sector [8]. While various data sets that represent students are in danger of being misused, its potential to initiate personalized education for every student is still in effect. In spite threats, positive outcomes can be achieved if compliance to legal procedures of data sharing and clear-cut policies are in place.

In the US context, the Family Educational Rights and Privacy Act or FERPA protects students and its family members by “prohibiting educational institutions from denying parents and eligible students certain rights with respect to student records”. It also provides for inspection of students education records and for amendment whenever the data is deemed inaccurate or misleading. Moreover, FERPA prohibits the access of third party entities to personally identifiable information of students, such as names, address, social security numbers date of birth and other information linked to a student that identifies them “with reasonable certainty”, without parents’ consent. To date, FERPA and its implementing rules by the US Department of Education remains to be a regulatory scheme for protection of data held by public schools [9].

In the European context, on the other hand, the General Data Protection Regulation or GDPR addresses concerns that shall bring cultural shift in the way European businesses and institution handle personal data [10]. This major legislative move is seen as a crucial step in promoting fundamental rights of its citizens by enabling them to object to certain data processing and have control over the use of their personal data. Major changes is also seen in Europe’s education system as more accountability is passed on to schools in the way records are handled. As such, GDPR may require schools of having documented understanding of why and how the information is held and collected, when will data be deleted and who can access it. The General Data Protection Regulation replaces Data Protection Act and will take effect starting May 2018.

1.2.2 Republic Act No. 10173

In the Philippine setting, initiatives on data privacy was formalized when the Republic Act 10173 or Data Privacy Act of 2012 was approved on August 15, 2012 while its implementing rules and regulations was put into effect on August 24, 2016 . The law aims for the protection of Filipino’s personal information that resides from information and communications systems on both public and private sectors. It is divided into five parts; definitions and general provisions, the National Privacy Commission, rights of data subjects and obligations of personal information controllers and processors, provisions specific to the government and finally, penalties. In addition, the law provides for the protection of individual’s privacy while maintaining free flow of information thereby promoting innovation and growth, regulation of data collection, recording, organization, storage, modification, retrieval, consultation, use, consolidation, obstruction and destruction of personal data and the compliance of the country with international standards for data protection through the National Privacy Commission. Moreover the creation of this law

was patterned to the General Data Protection Regulation (GDPR) of Europe [11].

1.2.3 Republic Act No. 8792

The Republic Act No. 8792 , otherwise known as the E-Commerce Act of 2000 in the Philippines shall “facilitate domestic and international dealings, transactions, arrangements, agreements, contracts and exchanges and storage of information through the utilization of electronic, optical and similar medium, mode, instrumentality and technology to recognize the authenticity and reliability of electronic documents related to such activities and to promote the universal use of electronic transaction in the government and general public. Moreover, the act applies to any kind of data message and electronic document used in the context of commercial and non-commercial activities to include domestic and international dealings, transactions, arrangements, agreements, contracts and exchanges and storage of information” [11], [12].

This act plays an important role in terms of compliance with RA 10173 as it outlines data processing initiatives of both public and private sectors in order to create an information-friendly environment which supports and ensures the availability, diversity and affordability of ICT products and services. As such, protection of data becomes imperative.

1.2.4 NPC and the Data Privacy Accountability and Compliance Framework

The promulgation of RA 10173 paved the way for the establishment of the National Privacy Commission under the Department of Information and Communications Technology. The said commission shall oversee and enforce the provisions of the act thereby ensuring compliance of personal information controllers in both public and private sectors and upholding their right to privacy. The NPC also provides for the Data Privacy Accountability and Compliance Framework as stipulated in the commission’s toolkit. The framework is composed of 5 pillars; appointment of Data Protection Officer (DPO), conduct of a Privacy Impact Assessment (PIA), Development of Privacy Management Program and Privacy Manual (PMP&PM), Implementation of Privacy and Data Protection Measures (PDPM) and finally, development of Breach Reporting Procedures (BRP) [3]. The framework is shown in Figure 1.

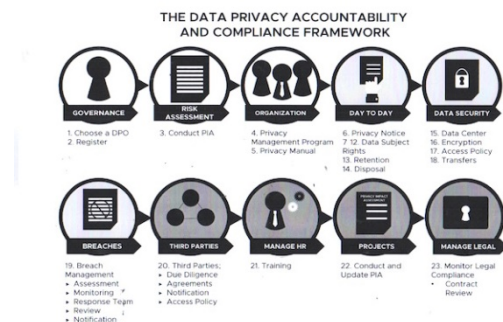


Figure 1. Data Privacy Accountability and Compliance Framework

1.3 Research Questions and Objectives of the Study

This study aims to assess the level of compliance of SUCs with RA 10173 thereby determining the challenges and issues they

faced in abundance with the law. The main research question revolves around how and why will SUCs comply with the RA 10173 and how are they integrating and applying the said law in their information systems and academic processes. In light of the objectives and main research question, the following specific research questions are considered:

1. *What are the data processes and information systems that deal with personal data?*
2. *How does SUCs compliance with RA8792 affect its compliance with RA 10173?*
3. *What are the documents submitted for the initial deadline of NPC regarding compliance with RA 10173?*
4. *How do determinants of compliance by Sutinen and Kuperan's [13] affect SUCs compliance with RA 10173?*
5. *What information security controls and strategies do SUCs apply in compliance with RA 10173?*
6. *What changes were made from the previous Information Systems Strategic Plan (ISSP) compared to the current?*

1.4 Scope of the Study

The SUC under analysis in this study is Mindanao State University- General Santos City (MSU-GSC) through their computer center director, Professor Rodulfo L. Dorado. MSU-GSC is a premiere regional university in Region 12, Mindanao that has a special mandate on integration of different cultural communities into the mainstream socio-cultural and political life by providing them opportunities in terms of quality and timely education essential for their development thereby producing skilled manpower contributing to the economic development of its area.

2. METHODOLOGY

The study employs a qualitative single holistic case study in order to explain the link between the identified phenomenon and the context where it exists. Yin [7] posits that a qualitative case study is a form of empirical inquiry that investigates a given phenomenon through multiple sources of evidence. The case study framework used is shown in Figure 2.

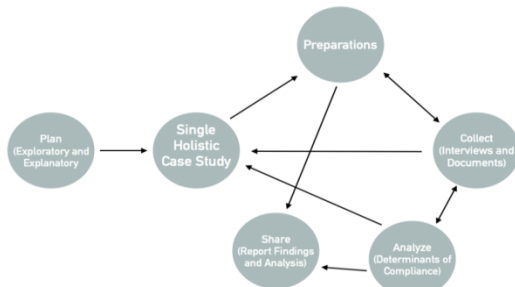


Figure 2. The Case Study Model

The process starts with a planning phase where research questions were formulated. The questions were exploratory and explanatory nature as they are in the context of how and why respectively as explained by Yin [6], [7]. The design phase is represented by the single holistic case study as this paper focused on a single and significant unit of analysis, which is Mindanao State University-General Santos City and its compliance initiatives. Preparations come by defining case study protocols in the form of consent letters to the institution, through the focal person, and for

documents needed to be reviewed. Series of interviews were done after the approval of the request to conduct the case study. Data from series of interviews, email correspondence and document reviews were then collected. Examination of the institution's Information Systems Strategic Plan (ISSP), which is considered as the primary document to be assessed for RA 10173 compliance, were also made. Analysis of data is done by applying the determinants of compliance by Sutinen and Kuperan [13]. As case studies in general can validate existing theories, this study was able to narrow down the list of determinants from the original proposition by analyzing interview narratives as shown in Figure 3.

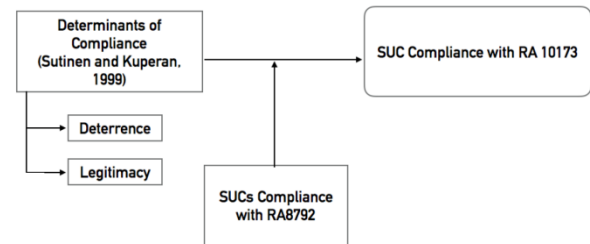


Figure 3. Operational Framework

In addition, the moderating effect of compliance with RA 8792 was included in order to determine its impact on SUCs compliance with RA 10173 as the former lays down all data processing initiatives [12] thereby determining what information systems are to be included in the assessment of compliance by the university. Pattern matching technique of Trochim [14], as seen in Figure 4, was also applied where predicted patterns are compared with empirical patterns [11], [14]. Consequently, this study has identified three patterns in determining compliance status of MSU-GSC with RA 10173 as presented in Table 1.



Figure 4. Pattern Matching Process

It can be gleaned from Table 1 that in order for SUCs to be fully compliant, the match between the ISSP document and the 5 pillars of data privacy accountability and compliance framework must both be High as reflected in Pattern A. Pattern B shows a High remark for ISSP while Medium to High (<5) resulting to Partially Compliant result. Finally, a Low remark for both ISSP and 5 pillars as reflected in Pattern C results to being not compliant.

Table 1. Compliance Patterns

Pattern	ISSP	Five Pillars	Match
Pattern A	High	High (5/5)	Fully Compliant
Pattern B	High	Medium to High (<5)	Partially Compliant
Pattern C	Low	Low	Not Compliant

Finally, the share phase is expected to be done once the study is completed and the results will take a significant role in answering the questions of this study in light of academic inquiry.

3. RESULTS AND DISCUSSIONS

Based on the series of interviews conducted with Professor Rodulfo Dorado of the MSU-GSC computer center, the university

is found to have continuously exerted efforts in maintaining information systems necessary for operations. The primary focus of the computer center, as of this writing, is on the school's management system and internet use. These two were prioritized so as not to hamper enrolment procedures and academic activities needing internet connection such as library services, computer laboratory classes and grade query transactions done online through its website in addition to activation of LDAP servers for its directory services.

Table 2. List of Information Systems from ISSP 2015-2017

MINDANAO STATE UNIVERSITY- GENERAL SANTOS CITY (MSU-GSC)					Presence of Personal Data		
	Information Systems ISSP 2015-2017	Owner	Status	Description	Y/N	Internal Processor	External Processor
1	Schools Management System	Registrar	Being Developed	Integrates key activities from admission, course scheduling, registration/enrollment, assessment, billing, cashing, and students records management.	Y	All personnel handling student data	CHED, CDA
2	MSU- General Santos Web Portal	OC	Being Developed	A content Management System that will adhere to the CHED template for HEIs will be adopted. CHED's content management module will be largely followed and enhanced by features unique to MSU-Gen. Santos.	N	MSU-GSC Stakeholders	CHED, Public
3	Admission Management System	Admission Office	For Development	A subsystem of the School Management System. A web-based portal that will be used to disseminate information for the annual admission and scholarship examination and electronically publish and communicate results to students.	Y	Admission Office	Incoming students, parents, guardians
4	Virtual Learning Management System	MSU-GSC	For Adoption	To develop this fully-developed system through Virtual Learning Management System (VIMS) that enables an academic institution to provide e-learning for students anytime and anywhere that addresses the issues of distance, digital divide and disaster mitigation, initiative of the government.	N	Academic Sector	All Learners
5	Property Management and Inventory Information System	SMD	For Development	The system will maintain an inventory of all property, materials and equipment and monitor the issuance of MRNs.	N	SMD	CDA, CHED
6	Executive Information System	OC	For Development	The system is designed to serve the information needs of management, hence will contain summaries of over-all operations.	Y	OC, VCAA, VCAF	CHED
7	Document Management System	OC	For Development	A web-based system for electronic creation, receiving and filing documents. It shall store and catalog digitized documents or soft-copy files for electronic access, retrieval and printing.	N	All employees	CHED
8	Teacher's Evaluation Report System	VCAA	Being Developed	The system archives and computes teacher's performance as rated by students, peers and supervisors.	Y	VCAA	VCAA, VCAA Staff
9	Research and Extension Projects Tracking Information System	RDO	For Development	This is an integrated system that will maintain a database of research proposals, ongoing research and extension projects, completed projects and researchers, with Web interface.	N	RDO, Academic Sector	General Public
10	Student Portal	OSA	For Development	The system will serve as a single stop shop for the students to verify their academic status and academic related concerns.	Y	Students, Faculty Guidance Office, OSA	Parents and Guardians
11	Financial Information System	MSU-GSC	For Development	This system will integrate all finance-related functions such as budgeting, accounting, cashing and purchasing and generate both soft and hard copies of reports.	N	Budget, Accounting, Cashier	CDA, DBM
12	Human Resource Information System	HRMO	For Enhancement	The system will store and retrieve records of all personnel including personal information, trainings, experience, leaves and absences, performance ratings, etc.	Y	HRMO, All Employees	-

Information systems present in the university are still used and are still in place as security and privacy measures are always part of its planning, development or adoption although limited to firewalls and built-in server security measures. Looking at the ISSP document, it can be learned that there is not much significant difference with the twelve information systems for the year 2015 to 2017 as compared to only ten for the year 2018-2020. The interview results explained that while there are systems that are yet to be developed, most of the existing systems are with the status for enhancement, adoption or being developed and were carried over from the previous to the new ISSP. Tables 2 and 3 show data processes and information systems where personal data resides as well as the comparison as to the changes made from the previous up to the current ISSP.

It was also found out, through observations, that not all organizational processes are automated leading to partial compliance with RA 8792. However, continuous efforts are made by computer center unit as it has its own means of integrating newly developed systems thereby eliminating the need to re-engineer the whole existing processes. Consequently, compliance with RA 10173 is not hampered in any way but is more encouraged so as data protection measures will still be pursued in future developments.

To explore the level of compliance of MSU-GSC with RA 10173, the pattern matching technique was applied using the information gathered and documents collected. It can be drawn from Table 4 that before the enactment of RA 10173 and before NPC took action, no compliance is observed thereby resulting to Pattern C. However, during the enactment of RA 10173 and after NPC took action, it can be observed that both Documents, Archival Records

and Interviews got the remark of Pattern B or partially compliant. This can be attributed to their initial compliance of appointing a temporary data privacy officer while registering the university's data processing systems to NPC are also being processed.

Interestingly, the evidence Privacy Impact Assessment (PIA) earned different remarks. Results of the interview revealed that while the computer center is aware of privacy concerns, there is a need to document their plans and measures to formalize it. As such, although it is not reflected on the documents, the interview revealed otherwise causing both pillar to have different remarks. Moreover, Professor Dorado discussed that the Philippine Association of State Universities and Colleges (PASUC) has already worked on a privacy manual that shall cover pillars 2 to 4 and is up for final revision As such, MSU-GSC is also anticipating the distribution of the said manual for them to develop privacy measures along the line of PASUC standards. Overall, after employing pattern matching, MSU-GSC was deemed Partially Compliant.

Table 3. List of Information Systems from ISSP 2018-2020

MINDANAO STATE UNIVERSITY- GENERAL SANTOS CITY (MSU-GSC)					Presence of Personal Data		
	Information Systems ISP 2018-2020	Owner	Status	Description	Y/N	Internal Processor	External Processor
1	Schools Management System	Registrar	Being Developed	Integrates key activities from admission, course scheduling, registration/enrollment, assessment, billing, cashing, and students records management.	Y	All personnel handling student data	CHED, CDA
2	MSU- General Santos Web Portal	OC	Being Developed	A content Management System that will adhere to the CHED template for HEIs will be adopted. CHED's content management module will be largely followed and enhanced by features unique to MSU-Gen. Santos.	N	MSU-GSC Stakeholders	CHED, Public
3	Admission Management System	Admission Office	For Development	A subsystem of the School Management System. A web-based portal that will be used to disseminate information for the annual admission and scholarship examination and electronically	Y	Admission Office	Incoming students, parents, guardians
4	Virtual Learning Management System	MSU-GSC	For Adoption	To develop this fully-developed system through Virtual Learning Management System (VIMS) that enables an academic institution to provide e-learning for students anytime and anywhere that addresses the issues of distance, digital divide and disaster mitigation, initiative of the government.	N	Academic Sector	All Learners
5	Executive Information System	OC	For Development	The system is designed to serve the information needs of management, hence will contain summaries of over-all operations.	Y	OC,VCAA,VCAF	CHED
6	Document Management System	OC	For Development	A web-based system for electronic creation, receiving and filing documents. It shall store and catalog digitized documents or soft-copy files for electronic access, retrieval and printing.	N	All employees	CHED
7	Teacher's Evaluation Report System	VCAA	Being Developed	The system archives and computes teacher's performance as rated by students, peers and supervisors.	Y	VCAA	VCAA, VCAA Staff
8	Research and Extension Projects	RDO	For Development	This is an integrated system that will maintain a database of research proposals, on-going research and extension projects, completed projects and researchers, with Web interface.	N	RDO, Academic	General Public
9	Student Portal	OSA	For Development	The system will serve as a single stop shop for the students to verify their academic status and academic related concerns.	Y	Students, Faculty Guidance	Parents and Guardians
10	Equipment Procurement and Management Information System	SMD	Being Developed	The system will maintain an inventory of all property, materials and equipment and monitor the issuance of MRNs.	N	SMD	COA, CHED

When asked about determinants of compliance of the university to RA 10173, Professor Dorado discussed about deterrents and legitimacy as primary reasons. Accordingly, MSU- GSC will have to comply because it is the law. While the motive behind the formulation of the law is the sense of rightness and justice in protecting private data, which is morally upright in nature, whether it be morally right or wrong becomes immaterial since the determining factor for compliance is the law and not one's conscience.

Table 4. Pattern Matching Results

		Expected Patterns		Pattern A Pattern B Pattern C		ISSP High High Low		3 Pillars High (1-3) High (1-3) Low		Match Fully Compliant Partially Compliant Not Compliant					
Case	Source	Event	Evidence	Rating	Evidence	Rating	Evidence	Rating	Evidence	Rating	Evidence	Rating	Match		
Documents Archival Records		Before Enactment of R.A. 10173	ISSP	Low	DPO	n/a	PIA	n/a	PMP and PM	n/a	PDPM	n/a	BRP	n/a	Pattern C
		Before NPC Took Action	ISSP	Low	DPO	n/a	PIA	n/a	PMP and PM	n/a	PDPM	n/a	BRP	n/a	Pattern C
		During Enactment of R.A. 10173 and After NPC Took Action	ISSP	High	DPO	High	PIA	Low	PMP and PM	Low	PDPM	Low	BRP	n/a	Pattern B
		After NPC Took Action	Interviews	ISSP	High	DPO	High	PIA	Medium	PMP and PM	Low	PDPM	Low	BRP	n/a
Legend: DPO Data Privacy Officer PIA Privacy Impact Assessment PMP and PM Privacy Management Program and Privacy Manual PDPM Privacy and Data Protection Measures BRP Breach Reporting Procedures															

As a result, deterrence, being the most common factor, and legitimacy were identified as constructs that has a significant

causal effect on the compliance behavior. Deterrence is defined as a strategy where punishment is enforced, surpassing any economic gain from failing to cope with requirements, to promote industry compliance while legitimacy is defined as one's recognition or acceptance of a governing body [6]. Other constructs were neglected, based on the results, as they did not show any causal relationship needed in understanding the phenomenon being analyzed in the study.

Interview narratives also revealed three factors that contribute primarily on the challenges on compliance by the university; lack of better understanding, budgetary issues and time constraints. Lack of better understanding emanates from not getting everyone involved in the organization on the initiative of protecting university data from the highest authority going across all units. In addition, the role of the DPO might also be too wide to absorb as greater understanding of its roles and accountabilities is needed for everyone to maintain compliance initiatives. Budgetary constraints, on the other hand, is always considered an issue in any government agency as IT investments may be overshadowed by other priorities or projects as these investments can be very expensive. However, with the existence of the law, the university will be more encouraged to include it for the coming allocation. Finally, in spite being promulgated on the year 2012, the NPC's awareness campaign came only in 2016. As such, the impact of the law and the severity of its punishment was made known late for universities to take action immediately as the compliance window was given for only a year, making time limited.

4. CONCLUSION

This paper contributes to the determination of the level of compliance of government agencies, from a State University perspective, and understanding of the challenges they face in order to abide by the law in RA 10173. MSU-GSC was found out to be partially compliant and their reason for compliance is due to deterrence and legitimacy. In spite challenges, the university continues to exert effort with their compliance initiatives thereby pursuing what is right and ethical in the field of security and privacy. Moreover, the study may serve as a call for the National Privacy Commission to intensify its efforts in extending support to SUCs by providing dialogues in every region concerning RA 10173 for greater appreciation of the law leading to more enthusiastic compliance performance.

5. FUTURE WORK

As this study uncovers compliance challenges faced by SUCs, a large area is left for discovery in future researches like the issue of RA 10173 compliance in government sectors, specifically SUCs, by studying compliance of inter-governmental organizations and identifying critical success factors and best practices in implementing an effective data protection policies along the line of data privacy act.

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